

EXAM 7 – SPRING 2012

18. (3 points)

Two insurance companies want to select appropriate risk measures, based on their aggregate loss distributions, for capital allocation purposes. The four risk measures under consideration are:

Standard deviation
Semi-standard deviation
VaR
TVaR

a. (1.5 points)

Company A writes homeowners and commercial property insurance in several coastal states. Its main concern is hurricane risk. Assess the appropriateness of each risk measure for Company A and determine the best one.

b. (1.5 points)

Company B writes personal auto insurance in several mid-western states. Its main concern is the price adequacy of their book. Assess the appropriateness of each risk measure for Company B and determine the best one.

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Question 18 Sample Answer

Solution 1

- a) Standard deviation: Not appropriate since it doesn't work well with skewed distributions like lines subject to hurricane losses

Semi standard deviation: Not appropriate since it will end up ignoring part of the distribution, and won't highlight hurricane risk enough (since it's total loss)

VaR – Not appropriate since its only one point, VaR(95) may not be high enough to capture hurricane risks.

T-VaR – While not ideal (I would use distortion of RTVaR) it's the best one in the list because it will average all loss values above VaR, better capturing hurricane risk.

- b) Standard deviation: Not appropriate here because it builds in positive or beneficial deviations

Semi-standard: Best one since it uses the whole distribution, and focuses only on negative outcomes, even the small ones.

VaR an T-VaR: Not appropriate for two reasons

- Estimating at high levels ignores too much risk
- Estimating at low levels puts too little weight on high risk (tail events)

Also, you would really want to look at the whole distribution here

Solution 2

- a) The standard deviation and semi-standard deviation are mean based risk measures they would not be appropriate in this case because for hurricane risk you are more concerned with the tail. As hurricane losses might be large but infrequent.

VaR and TVaR would be more appropriate as they are tail risk measures. TVaR might be even more applicable because it measures the average loss in the tail not just the probability of experiencing a loss above a certain threshold like the VaR.

- b) To asses the risk of price inadequacy a mean-based risk measure would be more appropriate because it takes into account the entire loss distribution. Therefore I would use either the standard deviation or the semi-standard deviation. Semi-standard might be slightly more appropriate as it focuses on only adverse outcomes which is what we are concerned about.

Examiner Comment

Most candidates did fairly well at assessing the risk measures, although oftentimes they defaulted to defining the measure as opposed to truly assessing its value for the given insurer. Semi-standard deviation was the most misunderstood of the four risk measures. A selection of the most appropriate risk measure for each situation was also required. A number of candidates did not do this. Of those candidates who did correctly identify the best measure, some failed to explain why that was indeed the best measure for the given situation.